VF Asia Footwear Laboratory Testing Manual

Test Method for Shrinkage	Test method # VF- E13 Page 1 of 3
Date Issued : March 14, 2016 **Revision # : 2** **Reason for Revisions : Temperature for expanded components such as EVA , sponge rubber lowered from 70 ° C to 60 ° C**	**Created by : Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the amount of shrinkage which a material undergoes as a result of exposure to heat for a specified time period.

REFERENCE: BS 5131

EQUIPMENT AND MATERIAL:

1. Air-circulating oven, capable of maintaining a temperature of 60 ± 1ºC and 70 ± 1ºC
2. Steel Scale, capable of measuring to 0.5mm.
3. Vernier caliper
4. Foam , Midsole, Die Cut EVA , Unit-sole/ Outsole and Sock-liner

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) For MD / IM midsoles , Pre-formed Sock-liners , unit-soles , outsoles

Note: Test plaques cannot be used to determine shrinkage for MD/ IM midsole, Pre-formed sock-liners, unit-soles and outsoles except for Development sample.

1 Draw a line from point A to point B. Line AB should be located approx. 1 cm from the front edge of midsole or unit-sole. Chose the widest width.
2 Draw a line from point C to Point D. line CD should be located approx. 1 cm from the back edge of the midsole or unit-sole. Chose the widest width.
3 Draw a line from point E to point F, Line EF should intersect both line AB and line CD.

Note: For EVA / MD / Foam / Sponge Rubber, if the thickness on the front edge is less than 7 mm, should adjust the location towards the back – edge. Meaning ensure that the thickness for shrinkage measurement should be at least 7.00 mm.

B) For Sock-liners, Die Cut EVA, Foam, Test plaque

1. Die or hand cut samples which are at least 12.0cm long and have a width of 2.54cm.
2. Draw a line (L) which is 10.0cm long along each of the test samples .line L should be drawn one cm from each edge of the test sample.

Note: For EVA / MD / Foam / Sponge Rubber, thickness of test plaque should be at 10 ± 2 mm.

REPETITIONS: Test at least two (2) samples for A and three (3) samples for B.

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TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) For MD / IM midsoles, Pre-formed Sock-liners, unit-soles, outsoles

1. Check oven calibration, oven temperature must be :
   - Expanded components such as MD / IM / Sock- liners, expanded unit-soles or sponge rubber - test temperature at **60.0 ± 1.0ºC.**
   - Solid unit soles and outsoles - test temperature at **70.0 ± 1.0ºC.**
2. Use a Vernier caliper but if it is not long enough, could use a steel scale. Ensure that the steel scale is stiff enough that it will not bend when taking a length measurement.

   Note: Length measurements must be taken using the bottom edge of the steel scale –to prevent it from bending.
3. Place the bottom edge of the steel scale on the midsole or unit-sole such that the steel scale contacts the foam at both point E and Point F of line L. The length **(L1)** is measured from the inside edge of lines AB and CD, Measure to the nearest 0.5mm.
   (Note : In some cases shrinkage for width should be measured also . Use lines AB for width #1 and CD for width #2)
4. Place the test sample in an oven at **60.0±1.0 ^{0}C or 70.0±1.0 ^{0}C** for 60+/-1 minutes.

Note: For Rubber Unit-soles / Outsoles add the following test parameters:
- **Test another sample in an oven at 70.0 ± 1.0 ^{0}C for 3 days (72 hrs)**
- **Test another sample in hydrolysis oven at 60.0 ± 1.0 ^{0}C, 95% RH for 7 days.**

5. Remove the test sample from the oven, and allow it to cool for at least 30 minutes at 23.0+/-2. 0 ^{0}C.
6. Using a Vernier caliper or steel scale, re-measure the length **(L2)** to the nearest 0.5mm in the same manner as described in Step 2-3.

B) Die – Cut EVA, and test plaque for sock-liners, foam, expanded rubber or sponge rubber, Outsoles

1. Check oven calibration, oven temperature must be:
   - Die – Cut EVA, and test plaque for sock-liners, foam, expanded rubber or sponge rubber - test temperature at **60.0 ± 1.0ºC.**
   - Solid outsoles - test temperature at **70.0 ± 1.0ºC.**

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2. Using a Vernier caliper, measure the length of line L to the nearest 0.1mm **(L_1).**
3. Place the test sample in an oven at **60.0±1.0 °C or** 70.0 ± 1.0 °C for 60+/-1minutes.
4. Remove the test sample from the oven and allow it to cool for at least 30 minutes at 23.0±-2.0 °C.
5. With the use of a Vernier caliper, re-measure the length of line L to the nearest 0.1mm **(L_2)**

CALCULATION:

Shrinkage in the length direction (%) = $\frac{L1-L2}{L1}$ *100

REPORT: The higher of the two measurements to the nearest 0.1%.

